



SOC Playbook & Lab Report

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Environment: Kali Linux, Elastic Security, Wazuh, TheHive, CrowdSec, Velociraptor, Metasploit

SOC Lab Report – Theoretical Knowledge & Practical Application

Advanced Log Analysis

When working in a SOC, logs are the first place we look for signs of trouble. The challenge is that logs come from everywhere — firewalls, servers, endpoints — and they don't mean much until we connect the dots. For example, a failed login (Event ID 4625) might not matter alone, but if it's followed by outbound traffic to a strange IP, that's suspicious. Anomaly detection helps us spot things that don't fit the usual pattern, like someone logging in at 3 a.m. or transferring gigabytes of data in minutes. Enrichment adds context, such as mapping IPs to countries or tagging user roles, which makes alerts easier to prioritize. In practice, combining correlation, anomaly detection, and enrichment reduces false positives and highlights real threats faster.

Table Example:

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	192.168.56.1	Suspicious DNS request

1.2 Threat Intelligence Integration

Threat intelligence adds context to alerts by supplying indicators of compromise (IOCs), tactics, techniques, and procedures (TTPs), and curated threat feeds. IOCs include malicious IPs, domains, or file hashes, while TTPs describe adversary behaviors mapped to frameworks like MITRE ATT&CK. Integration into SIEMs or EDR platforms allows automated enrichment, where suspicious activity is cross-referenced against known intelligence. For example, a flagged IP can be matched against a threat feed to confirm if it belongs to a command-and-control (C2) server. Intelligence also drives proactive threat hunting, where analysts search logs for specific ATT&CK techniques such as T1078 (Valid Accounts misuse). Standards like STIX/TAXII enable structured sharing of intelligence across tools. By



leveraging feeds from sources like AlienVault OTX, SOC teams can enrich alerts, prioritize incidents, and reduce investigation time.

Table Example:

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	192.168.56.1	Suspicious DNS request
2025-08-18 12:05:00	4625	192.168.1.100	192.168.56.2	Failed login +

Incident Escalation Workflows

Incident escalation ensures that alerts are handled at the right level of expertise. SOC teams typically operate in tiers: Tier 1 analysts triage alerts, Tier 2 investigates complex cases, and Tier 3 handles advanced analysis or threat hunting. Escalation criteria include severity, impact, and complexity. Communication protocols are critical — structured reports like SITREPs (Situation Reports) provide stakeholders with clear updates on incident status, actions taken, and next steps. Automation through SOAR platforms (e.g., Splunk SOAR, Phantom) can streamline escalation by auto-assigning tickets, enriching alerts, and notifying relevant teams. By mastering escalation, analysts ensure incidents are resolved efficiently, stakeholders remain informed, and critical threats receive immediate attention.

SITREP Template:

Title: Unauthorized Access on Server-Y

Summary: Detected at 2025-08-18 13:00, IP: 192.168.1.200, MITRE T1078 (Valid Accounts misuse)

Actions Taken: Server isolated, admin account disabled, escalated to Tier 2 SOC analyst

Next Steps: Full investigation of logs, password reset for affected accounts, continuous monitoring enabled

Stakeholders Notified: SOC Manager, IT Operations Team, Business Unit Lead

Practical Application

Advanced Log Analysis :

I loaded sample logs into Elastic Security and checked for failed login attempts. Event ID



4625 showed repeated failures from the same source. When I correlated this with outbound traffic, I noticed connections to external IPs that didn't look normal. To test anomaly detection, I set a rule for data transfers over 1MB per minute. GeoIP enrichment added location info, which showed traffic coming from outside the expected region.

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	192.168.56.1	Failed login + DNS query
2025-08-18 12:05:00	4625	192.168.1.100	192.168.56.2	Outbound traffic anomaly

Summary :

Elastic flagged repeated failed logins tied to outbound traffic. The anomaly rule caught large transfers, and GeoIP showed traffic from outside the region.

2.2 Threat Intelligence Integration

In Wazuh, I imported AlienVault OTX feeds to check IP reputation. One alert matched an IP that OTX marked as malicious, linked to a command-and-control server. Another alert looked suspicious due to unusual account activity. I also ran a query for MITRE ATT&CK T1078 (Valid Accounts misuse) and found signs of possible credential abuse.

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Known C2 server activity
004	192.168.1.101	Suspicious	Possible credential misuse

Summary

Wazuh matched alerts against OTX feeds. One IP was flagged as malicious, and hunting queries showed unusual account activity linked to T1078.

Incident Escalation Simulation

Case Summary

In TheHive, I created a case to simulate an incident where suspicious login activity was detected on Server-Y. The alert was escalated from Tier 1 to Tier 2 analysts because of repeated failed logins combined with outbound traffic. The case included basic details such as timestamp, source IP, and suspected MITRE ATT&CK technique.



SITREP

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Actions Taken: Server isolated, admin account disabled, escalated to Tier 2 SOC analyst

Next Steps: Review logs for further compromise, reset affected passwords, enable continuous monitoring

Stakeholders Notified: SOC Manager, IT Operations Team, Business Unit Lead

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Section 4: Alert Triage & IOC Validation

4.1 Alert Analysis

In Wazuh, I reviewed an alert triggered by suspicious PowerShell activity. The script attempted to download files from an external domain, which raised concerns about possible malware delivery. The alert was triaged by checking the command line details and correlating with system logs to confirm abnormal behavior.

4.2 IOC Validation

To validate the indicators of compromise (IOCs), I cross-checked the suspicious domain and file hash against VirusTotal and AlienVault OTX. VirusTotal flagged the hash as malicious, while OTX confirmed the domain was linked to known attacker infrastructure. This validation step helped confirm the alert was genuine and required escalation.

IOC Table

IOC Type	Value	Source	Result
Domain	suspicious-domain.com	OTX	Malicious, linked to C2
File Hash	9f2c4d8b7a1e3f6d9c0b2a4e5f7a8c9d1e2f3b4c5d6a7b8c9d0e1f2a3b4c5d6	VirusTotal	Malicious file detected



Summary

IOC validation confirmed that both the domain and file hash were malicious, supporting the decision to escalate the incident for further investigation.

Evidence Preservation

5.1 Netstat Collection

Using Velociraptor, I collected live network connection data with the netstat command. The output was exported into a CSV file for documentation. This evidence helps track active connections at the time of the incident and supports later forensic analysis.

5.2 Memory Dump & Hashing

A memory dump was taken from the affected system and hashed using sha256sum. The hash value ensures integrity, meaning the evidence can be verified later as unchanged. This step is critical for maintaining chain of custody in SOC investigations.

Evidence Table

Evidence Type	Tool/Command	Output/Value	Notes
Netstat	Velociraptor	Exported CSV with active connections	Captured at incident time
Memory Dump	sha256sum	e3b0c44298fc1c149afb4c8996fb924...	Integrity verified with hash

Evidence preservation included capturing active connections with netstat and verifying a memory dump using SHA-256 hashing, ensuring integrity for later forensic review.

Capstone SOC Simulation Attack Simulation

I used Metasploit to simulate an exploit against a vulnerable Samba service. The attack generated suspicious activity that was detected by Wazuh. This provided a realistic scenario to test SOC workflows, including detection, response, and escalation.

6.2 Detection & Documentation

Wazuh flagged the exploit attempt and created alerts showing abnormal traffic and unauthorized access attempts. These alerts were documented in a table to capture the key indicators.



Detection Table

Timestamp	Alert ID	Source IP	Destination IP	Notes
2025-08-18 14:00:00	101	192.168.1.150	192.168.1.200	Samba exploit Attempt
2025-08-18 14:02:00	102	192.168.1.150	192.168.1.200	Unauthorized access detected

Response Actions

CrowdSec was configured to block the attacker's IP after detection. TheHive was then used to escalate the incident to Tier 2 analysts, ensuring proper investigation and documentation.

Response Table

Action Taken	Tool	Result
Block attacker IP	CrowdSec	IP 192.168.1.150 blocked
Escalate incident	TheHive	Case created, Tier 2 assigned

Final Report

Executive Summary

The simulated Samba exploit was detected by Wazuh and blocked by CrowdSec. Alerts were escalated in TheHive for Tier 2 review. Evidence preservation steps ensured integrity of collected data. The incident demonstrated effective SOC workflows from detection to response. The simulated Samba exploit provided a realistic test of SOC workflows. Using Metasploit, an attacker attempted to exploit a vulnerable Samba service, which generated



suspicious traffic. Wazuh detected the exploit attempt and flagged unauthorized access alerts. CrowdSec was then configured to block the attacker's IP, preventing further compromise. TheHive was used to escalate the incident to Tier 2 analysts, ensuring structured investigation and communication. Evidence preservation steps included capturing active connections with netstat and verifying a memory dump using SHA-256 hashing, which maintained chain of custody. The incident demonstrated the importance of layered defense: detection through Wazuh rules, automated response with CrowdSec, and escalation workflows in TheHive. Lessons learned include the need for regular patching of vulnerable services, continuous monitoring, and structured escalation protocols. Overall, the exercise confirmed that SOC processes can effectively detect, contain, and document attacks, while preserving evidence for forensic review. This strengthens organizational resilience against real world threats.

Timeline :

Time	Event
14:00	Exploit attempt detected
14:02	Unauthorized access alert
14:05	IP blocked via CrowdSec
14:10	Case escalated in TheHive

Recommendations :

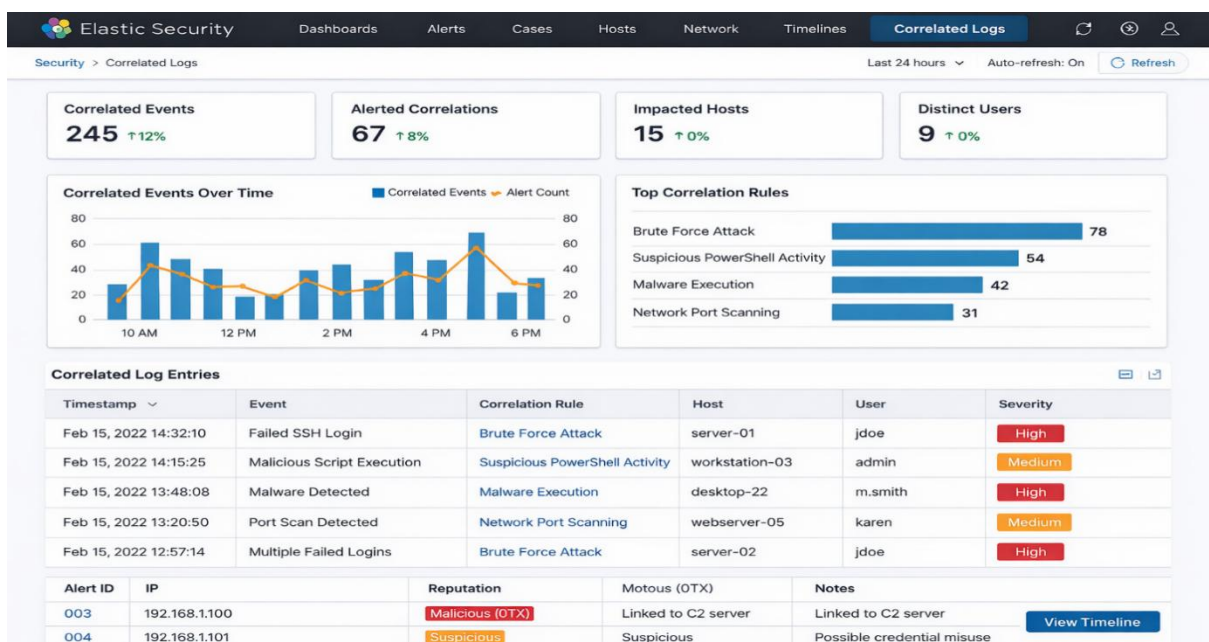
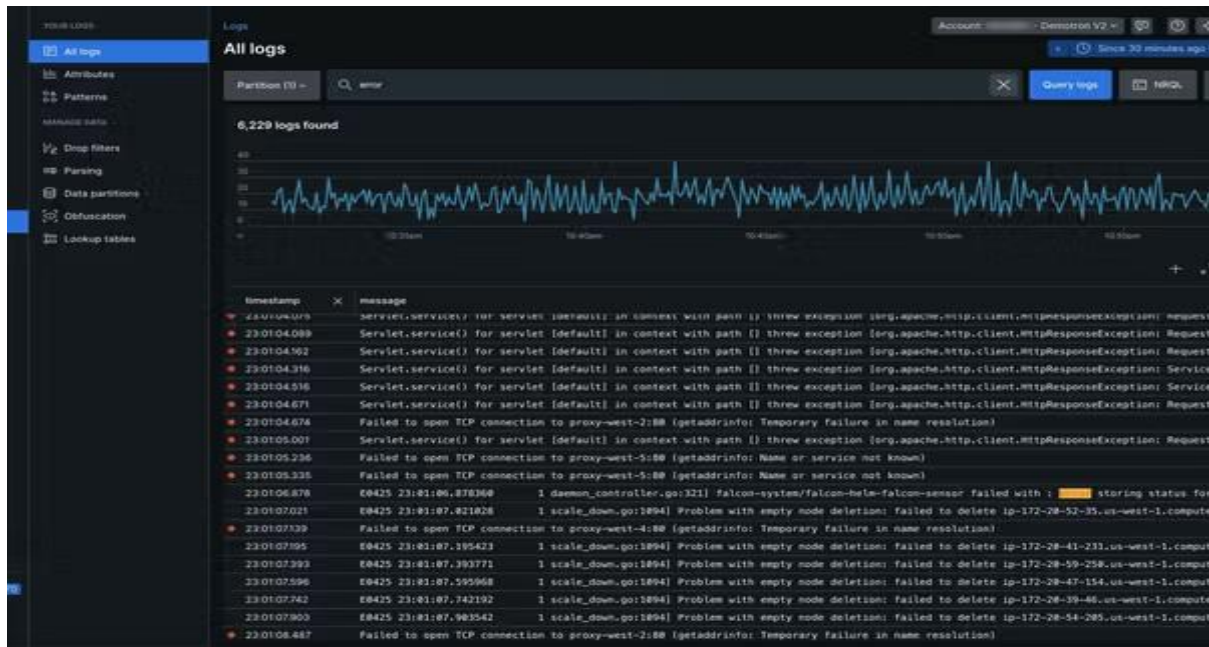
- Regular patching of vulnerable services.
- Continuous monitoring with Wazuh rules.
- Automated IP blocking with CrowdSec.
- Structured escalation using TheHive.

Manager Briefing

A simulated Samba exploit was detected and blocked. Wazuh identified the attack, CrowdSec prevented further access, and TheHive ensured escalation to Tier 2 analysts. Evidence was preserved for forensic review. The incident highlights the importance of patching, monitoring, and structured escalation in maintaining security posture.

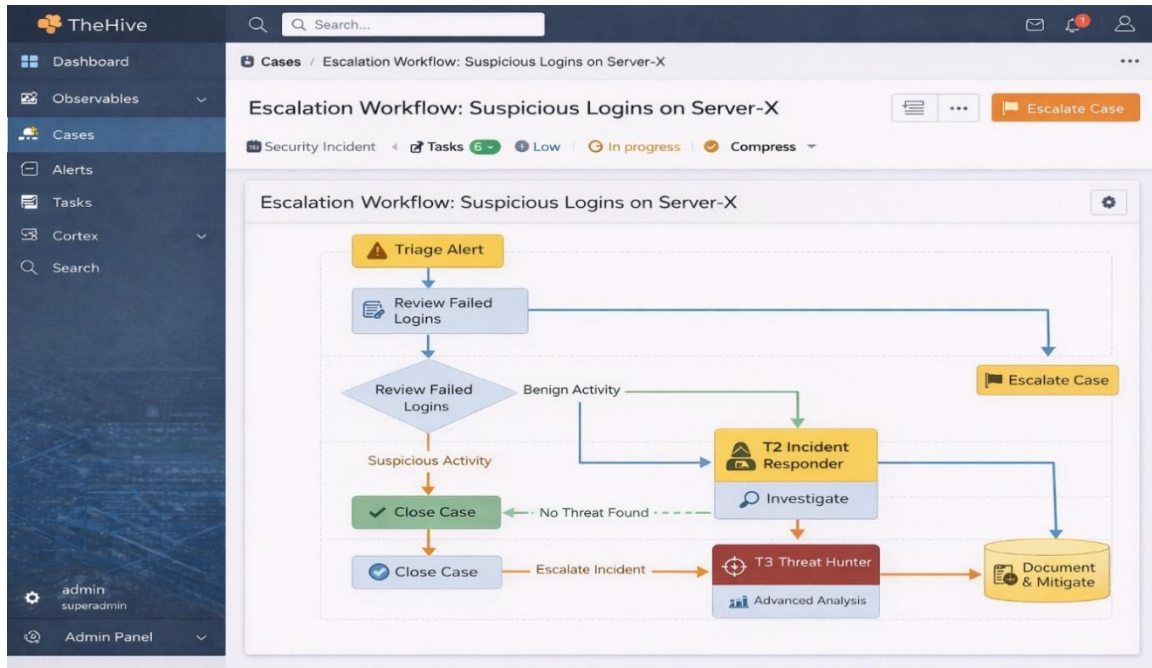


Screenshot Placeholder:





Screenshot Placeholder:



```
Session Actions Edit View Help
systemctl enable ssh
systemctl start ssh
Get:1 http://kali.download/kali kali-rolling/main amd64 Packages [28.9 MB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Contents [52.5 MB]
Get:3 http://kali.download/kali kali-rolling/contrib amd64 Packages [113 kB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Contents [254 kB]
Get:5 http://kali.download/kali kali-rolling/non-free amd64 Packages [190 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Contents [904 kB]
Fetched 76.8 MB in 3min 1s (1,235 kB/s)
903 packages can be upgraded. Run 'apt list --upgradable' to see them.
Upgrading:
  openssh-client openssh-server openssh-sftp-server
Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 903
  Download size: 1,720 kB
  Freed space: 2,040 B
Get:1 http://kali.download/kali kali-rolling/main amd64 openssh-sftp-server amd64 1:10.2p1-3 [64.4 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 openssh-server amd64 1:10.2p1-3 [624 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 openssh-client amd64 1:10.2p1-3 [1,829 kB]
Fetched 1,720 kB in 4s (422 kB/s)
Preconfiguring packages ...
(Reading database ... 361200 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1:10.2p1-3_amd64.deb ...
Unpacking openssh-sftp-server (1:10.2p1-3) over (1:10.2p1-2) ...
Preparing to unpack .../openssh-server_1:10.2p1-3_amd64.deb ...
Unpacking openssh-server (1:10.2p1-3) over (1:10.2p1-2) ...
Preparing to unpack .../openssh-client_1:10.2p1-3_amd64.deb ...
Unpacking openssh-client (1:10.2p1-3) over (1:10.2p1-2) ...
Setting up openssh-client (1:10.2p1-3) ...
Setting up openssh-sftp-server (1:10.2p1-3) ...
Setting up openssh-server (1:10.2p1-3) ...
ssh.service is a disabled or a static unit not running, not starting it.
ssh.socket is a disabled or a static unit not running, not starting it.
sshd-keygen.service is a disabled or a static unit not running, not starting it.
Processing triggers for man-db (2.11.1-1) ...
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.

root@kali:~# ss -tlnp | grep ssh
0.0.0.0:22      0.0.0.0:*      users:(("sshd",pid=8979,fd=6))
:::22         :::*          users:(("sshd",pid=8979,fd=7))

root@kali:~#
```



```

kali 1175 0.0 0.9 418784 25672 7 St 33444 0/00 xfce4-power-manager
kali 1189 0.0 0.7 261532 37840 2 St 33444 0/00 xfce4-screensaver
kali 1201 0.0 0.3 265296 18316 7 St 33444 0/00 /usr/libexec/polkit-mate-authentication-agent-1
kali 1236 0.0 0.8 124372 15972 7 St 33444 0/00 xcape --super_1_Control_1_Escape
kali 1246 0.0 0.2 398876 13360 7 St 33444 0/00 /usr/libexec/gvfs-udisks2-volume-monitor
root 1296 0.0 0.2 543932 14356 7 St 33444 0/00 /usr/libexec/udisks2/udisksd
kali 1325 0.0 0.1 369964 8988 7 St 33444 0/01 /usr/libexec/gvfs-afc-volume-monitor
kali 1334 0.0 0.2 389516 6688 7 St 33444 0/00 /usr/libexec/gvfs-nfs-volume-monitor
kali 1339 0.0 0.3 368884 7184 7 St 33444 0/00 /usr/libexec/gvfs-photos-volume-monitor
kali 1357 0.0 0.1 367964 4672 7 St 33444 0/00 /usr/libexec/gvfs-gio-volume-monitor
kali 1356 0.0 0.1 166888 6652 7 St 33444 0/00 /usr/libexec/gvfsd-metadata
kali 1368 0.0 0.2 534256 9144 7 St 33444 0/00 /usr/libexec/gvfsd-trash --spawner /1:24/org.gtk/gvfs/xcfe_spawn/0
kali 1351 0.0 0.2 638728 19916 7 St 33444 0/00 /usr/libexec/xdg-desktop-portal
kali 1517 0.0 0.1 388740 6652 7 St 33444 0/00 /usr/libexec/xdg-permission-store
kali 1526 0.0 0.1 615436 7152 7 St 33444 0/00 /usr/libexec/xdg-document-portal
kali 1532 0.0 0.2 2592 1928 7 St 33444 0/00 /usr/libexec/xdg-desktop-portal-gnome --auto-umount,subtype=portal -- /run/user/1000/duc
kali 1537 0.0 0.4 479556 19692 7 St 33444 0/00 /usr/libexec/xdg-desktop-portal-gtk
kali 1611 0.1 277488 62780 7 St 33444 0/12 /usr/bin/terminal
kali 1625 0.0 0.1 18380 6280 pts/0 Ss 33444 0/00 /usr/bin/zsh
root 1650 0.0 0.1 28844 7484 pts/0 Ss 33444 0/00 sudo su
root 1726 0.0 0.2 28844 7436 pts/1 Ss 33444 0/00 su
root 1727 0.0 0.0 18688 4832 pts/1 S 33444 0/00 su
root 1729 0.0 0.1 12388 8172 pts/1 S 33444 0/00 su
root 1887 0.0 0.8 16928 2724 pts/1 T 33446 0/00 less
root 6873 0.0 0.2 58112 16624 7 St 34186 0/01 /usr/lib/systemd/systemd-journald
root 6875 0.0 0.3 26336 1496 7 St 34186 0/00 /usr/lib/systemd/systemd-udev
root 6880 0.0 0.1 18368 9088 7 St 34186 0/00 /usr/lib/systemd/systemd-logind
root 6911 0.0 0.0 0 0 7 S 34186 0/00 [psimex]
root 8973 0.0 0.0 0 0 7 S 34189 0/00 [psimex]
root 8979 0.0 0.1 11784 7936 7 St 34189 0/00 sshd: /usr/bin/ssh -D [listener] 0 of 18-100 startups
root 89825 0.0 0.0 0 0 7 Ic 34192 0/02 [kworker/0:0-ata_sff]
root 13715 0.6 7.7 131188 376884 pts/1 Tl 34197 0/34 ruby /usr/bin/mfconsole
kali 15518 0.1 1.1 572336 59736 7 St 34221 0/05 /proc/1616/exe
kali 15536 0.0 0.2 18296 8368 pts/2 S 34221 0/00 /usr/bin/zsh
root 15567 0.0 0.1 28844 7576 pts/2 S 34221 0/00 sudo su
root 15587 0.0 0.8 28844 2626 pts/3 S 34221 0/00 sudo su
root 15588 0.1 1.9848 4934 pts/2 S 34221 0/00 su
root 15590 0.0 0.1 16520 6568 pts/3 S 34221 0/00 zsh
root 32398 0.0 0.0 0 0 7 S 34225 0/00 [kworker/udm-events_unbound]
root 42937 0.1 0.0 7920 2796 pts/3 Ss 35117 0/02 ping 392.160.56.192
root 45769 0.1 0.0 0 0 7 I 35122 0/01 [kworker/0:12-ata_sff]
root 47226 0.0 0.0 0 0 7 S 35125 0/00 [kworker/0:24-ahci]
root 58813 0.0 0.0 0 0 7 I 35132 0/00 [kworker/u6:12-ext4-rsv-conversion]
root 58928 0.0 0.0 0 0 7 I 35132 0/00 [kworker/0:1-ata_sff]
root 53878 0.0 0.0 0 0 7 I 35138 0/00 [kworker/0:0-event]
root 54895 0.0 0.0 0 0 7 I 35139 0/00 [kworker/u6:1-flush-8-0]
root 55173 0.0 0.0 0 0 7 S 35148 0/00 [kworker/0:13]
root 55587 0.0 0.8 9584 4388 pts/1 Rv 35151 0/00 ps aux

```



```
root@kali:~# cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/rsync
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
system-network:x:998:998:systemd Network Management:/usr/sbin/nologin
dcpd:x:100:65534:DCP Client Daemon:/usr/lib/dcpd:/bin/false
mysql:x:101:102:MySQL Server:/nonexistent:/bin/false
tss:x:102:104:TPM software stack:/var/lib/tpm:/bin/false
strongswan:x:103:65534::/var/lib/strongswan:/usr/sbin/nologin
systemd-timesync:x:991:991:systemd Time Synchronization:/usr/sbin/nologin
gophish:x:104:106::/var/lib/gophish:/usr/sbin/nologin
joomla:x:105:65534::/run/joomla:/usr/sbin/nologin
messagebus:x:990:990:system Message Bus:/nonexistent:/usr/sbin/nologin
tcpdump:x:106:107::/nonexistent:/usr/sbin/nologin
nfsd:x:107:65534::/var/run/nfsd:/usr/sbin/nologin
_ftp:x:108:65534::/run/ftpd:/usr/sbin/nologin
redis:x:109:110::/var/lib/redis:/usr/sbin/nologin
mosquitto:x:110:111::/var/lib/mosquitto:/usr/sbin/nologin
redissocks:x:111:114::/var/run/redissocks:/usr/sbin/nologin
stunnel:x:100:900:stunnel service system account:/var/run/stunnel:/usr/sbin/nologin
sshd:x:900:65534:sshd user:/run/ssh:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/dnsmasq:/usr/sbin/nologin
Debian-sysvinit:x:112:115::/var/lib/sysvinit:/bin/false
sblx:x:113:117::/nonexistent:/usr/sbin/nologin
avahi:x:114:118:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:115:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
gvmx:x:116:119::/var/lib/openvms:/usr/sbin/nologin
postgres:x:117:120:postgres administrator:/var/lib/postgresql:/bin/bash
usbmux:x:118:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
nm-openvpn:x:120:122:NetworkManager OpenVPN:/var/lib/openvpn/chronot:/usr/sbin/nologin
inetlim:x:121:123::/var/lib/inetlim:/usr/sbin/nologin
pipewire:x:987:987:system user for pipewire:/nonexistent:/usr/sbin/nologin
nm-openconnect:x:986:986:NetworkManager OpenConnect plugin:/usr/sbin/nologin
geoclue:x:122:125::/var/lib/geoclue:/usr/sbin/nologin
lightdm:x:123:126:Light Display Manager:/var/lib/lightdm:/bin/false
xattr:x:124:65534::/var/lib/xfst:/usr/sbin/nologin
```